

EE 210P: Digital System Design

Course Code	EE 210P
Course Name	Digital System Design
L-T-P-C	0-0-2-1
Prerequisites	Applied Electronics (IC161) and Applied Electronics Lab (IC161P).
Intended for	B. Tech (EE)
Distribution	Core for B. Tech (EE) & Elective for other UG programs.
Approval	32nd BoA (Dec 2019); 24th Senate (Feb 2020)
Replaces	EE 208P by the 32nd BoA (Dec 2019)

Experimental Modules for EE-210P

- Basics understanding of hardware descriptive language (Verilog / VHDL / System Verilog). Netlist synthesis, functional (Behavioral) & post-route simulations, and hardware prototyping on FPGA using latest design suites (like Xilinx ISE or Vivado or Model Sim etc.). [2 hours]
- Design and implementation of basic gates, multiplexers, de-multiplexers, various adders and multipliers in FPGA platform. [1 hours]
- FPGA implementations of flip-flops, registers, counters and circuitry verifying the register delays. [1 hours]
- Design and implementation of Processor ALU on FPGA. [2 hours]
- Design and implementations of random-access memory (RAM) and read only memory (ROM). These experiments must include the reading and writing operations of RAM for various applications. [1 hours]
- Understanding the static timing analysis of simple sequential circuits. It must demonstrate the two major states of any sequential circuits: timing violated and timing met states. [2 hours]
- Embedded experiments on LED and 7 -segment displays. [1 hours]
- Embedded experiment on Interfacing the LCD and LDR. [2 hours]
- Embedded experiments on interfacing servo motor with ultrasonic. [2 hours]

Text Books

1. Donald D. Givone, **Digital Principles and Design**, Tata McGraw-Hill Edition, 2012.
2. John M. Yarbrough, **Digital Logic (Applications and Design)**, Cengage Learning, 2011.

References

1. Dhananjay Gadre, **Programming and Customizing the AVR microcontroller**, Tata McGraw Hill, 2014.
2. Wayne Wolf, **FPGA based Systems Design**, Pearson Education, 2003.
3. Stephen Brown and Zvonko Vranesic, **Fundamentals of Digital with VHDL Design**, McGraw Hill, 2008.
4. Volnei A. Pedroni, **Circuit Design with VHDL**, The MIT Press, 2004.
5. Steve Kilts, **Advanced FPGA Design: Architecture, Implementation and Optimization**, J. Wiley and Sons, 2007.
6. Seetharaman Ramachandran, **Digital VLSI Systems Design**, Springer Verlag, 2012.
7. Peter J. Ashenden, **The designer's guide to VHDL**, Morgan Kaufmann, 2008.
8. Charles H. Roth Jr., **Digital Systems Design using VHDL**, Cengage Learning, 2014.